

Task 1 — Document 06: Workflow & User Flow Diagrams

Problem Statement ID: PS 26117 / SIH26117 **Title:** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work **Organization:** Mangalore Refinery and Petrochemicals Limited (MRPL) **Category:** Software | **Theme:** Smart Automation

1. High-Level User Flow Chart

Visual Diagram Flow: FLOWCHART TD	
User opens Workbench Web UI	→ ActionChoice{User Action}
ActionChoice	→ Upload File UploadType{File Format?}
UploadType	→ Extract Text via pypdf / File Read
UploadType	→ Send to GLM-OCR via Ollama
PDFProcess	→ Chunk Text into 500-char snippets
OCRProcess	→ Chunking
Chunking	→ Generate Vectors via nomic-embed-text
Embedding	→ Append to kb_index.json
UpdateKB	→ Show UI Notification: Chunks Indexed
ActionChoice	→ Enter Prompt CheckCoding{Coding Keywords Detected?}
CheckCoding	→ Route to qwen2.5-coder:7b
CodeRoute	→ Display Synthesized Code Response
CheckCoding	→ No CheckKnowledge{Knowledge Mode ON?}
CheckKnowledge	→ Route to qwen3.5:4b
GeneralRoute	→ Display General AI Answer

2. Detailed Sequence Diagrams

Sequence 1: Image Upload & GLM-OCR Ingestion Pipeline

Visual Diagram Flow: SEQUENCEDIAGRAM	
User → UI	Selects & uploads inspection_sheet.png
UI → API	POST /upload (File payload)
API → API	Detects image file extension (.png)
API → OCR	Send image payload for text extraction
OCR → API	Return extracted raw OCR text

API → API	Split OCR text into semantic chunks
API → Embed	Generate 768-dim vector embedding
Embed → API	Return vector array
API → Index	Append chunk text + vector + source filename
Index → API	Save updated index
API → UI	Return JSON success response (file, chunks count)
UI → User	Display notification: "inspection_sheet.png processed with GLM-OCR (3 chunks added)"

Sequence 2: Grounded RAG Query Execution (Knowledge ON)

■ Visual Diagram Flow: SEQUENCEDIAGRAM	
User → UI	Types query: "What should be checked during pump inspection?" (Knowledge ON)
UI → API	POST /chat { message, knowledge_enabled: true }
API → Embed	Embed user query string
Embed → API	Return query vector
API → Index	Load stored vectors & calculate cosine similarity
Index → API	Return top-5 context chunks (test_sop.txt, score: 0.9421)
API → LLM	Pass Grounded System Prompt + Top-5 Context + Query
LLM → API	Return factual answer based strictly on context
API → UI	Return JSON { answer, sources: [{ file: "test_sop.txt", score: 0.9421 }] }
UI → User	Render Grounded Answer + Source Pill Badges
API → UI	Return JSON { answer: "That information is not available in the knowledge base.", sources: [] }
UI → User	Render Anti-Hallucination Warning Message

Sequence 3: Code Generation Task Routing

■ Visual Diagram Flow: SEQUENCEDIAGRAM	
User → UI	Types prompt: "Write a python function to parse refinery sensor logs"
UI → API	POST /chat { message, knowledge_enabled: false }
API → Router	Analyze prompt keywords ("python", "function", "parse")
Router → API	Keyword match triggered -> Route: CODING
API → CoderLLM	Send prompt to qwen2.5-coder:7b
CoderLLM → API	Return synthesized Python code snippet

API → UI	Return JSON { route: "coding", model: "qwen2.5-coder:7b", answer: "```python..." }
UI → User	Render syntax-highlighted code block in UI